

Resolution is the process of replacing symbolic references from the type with direct references. This is done by searching the method area to locate the referenced entity.



Initialisation

During the Initialisation phase, all static variables are assigned their respective values as defined in the code, along with any values set in static blocks (if any are present). This process occurs in a top-to-bottom manner within a class and from parent to child in the class hierarchy. Typically, there are three types of class loaders:

1. **Bootstrap class loader:** Every JVM implementation must have a bootstrap class loader, which can load trusted classes. It loads core Java API classes located in the “`JAVA_HOME/jre/lib`” directory, commonly referred to as the bootstrap path. It is implemented in native languages such as C or C++.
2. **Extension class loader:** This class loader is a child of the bootstrap class loader. It loads classes found in the extensions directories such as “`JAVA_HOME/jre/lib/ext`” (extension path) or any other directory specified by the `java.ext.dirs` system property. It is implemented in Java by the `sun.misc.Launcher$ExtClassLoader` class.
3. **System/Application class loader:** This class loader is a child of the extension class loader. It is responsible for loading classes from the application class-